

Generating Query-Focused Summarization Datasets from Query-Free Summarization Datasets

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Abstract

Large-scale datasets are widely used to perform summarization tasks, but they may not include queries alongside documents and summaries. In the search for suitable datasets for Query-Focused Summarization (QFS), we identify two research questions: Is it possible to automatically generate evidence-based query keywords from query-free datasets? Does evidence-based query generation support the QFS task? This paper proposes an evidence-based model to generate queries from query-free datasets. To evaluate our model intrinsically, we compare the similarity between the original queries and the system-generated queries of two QFS datasets. We also perform summarization tasks using different pre-trained models, as well as a state-of-the-art (SOTA) QFS model, to measure the extrinsic performance of our query generation approach. Experimental results indicate that summaries generated using evidence-based queries achieve competitive ROUGE scores compared to those generated from the original queries.

1 Introduction

Query-focused summarization (QFS) focuses on generating summaries from original documents, where the summary is tailored to a specific given query. Researchers have implemented various neural models and proposed distinctive approaches that advance both extractive and abstractive query-focused summarization tasks (Lin, 2004; Gupta et al., 2007; Wan et al., 2007; Ouyang et al., 2011; Feigenblat et al., 2017; Nema et al., 2017; Haselqvist et al., 2017; Baumel et al., 2018; Abdullah and Chali, 2020; Xu and Lapata, 2020; Laskar et al., 2020; Su et al., 2021). However, the lack of appropriate datasets for QFS has always been a concern for researchers, as the unavailability of large-scale datasets makes the task more challenging (Fisher and Roark, 2006; See et al., 2017; Liu and Lapata, 2019; Abdullah and Chali, 2020).

Thus, the shortage of large-scale QFS datasets motivates the development of effective query generation approaches. To address this issue, we propose a context-oriented, evidence-based model that supports query-focused summarization by generating queries from documents in any query-free dataset. Using a transfer learning approach, we train our evidence-based model on the article, highlight pairs from the CNN/DailyMail dataset. To avoid data bias, we use different datasets, such as Debatepedia and TD-QFS, for the QFS task instead of CNN/DailyMail. Additionally, both datasets contain queries, allowing us to compare the performance of original queries with that of the generated evidence-based queries. Samples of the original and evidence-based queries from the TD-QFS dataset are shown in Table 1.

Sample Queries	
Original	Evidence-based
asthma causes	asthma chronic disease affects air
Lung cancer diagnosis	lung cancer screening currently routine practice

Table 1: Sample queries (original and the evidence-based) from TD-QFS dataset.

2 Related Work

Pre-trained models, including BERT (Devlin et al., 2019), GPT (Radford et al., 2019), RoBERTa (Liu et al., 2019), T5 (Raffel et al., 2020), LED (Beltagy et al., 2020), BART (Lewis et al., 2020), and PEGASUS (Zhang et al., 2020), have been widely used on various datasets such as Gigaword (Ma and Huang, 2006), CNN/DailyMail (Hermann et al., 2015), SQuAD (Rajpurkar et al., 2016), TD-QFS (Baumel et al., 2016), and Debatepedia (Nema et al., 2017) to perform summarization, machine translation, and other NLP tasks (Rush et al., 2015; Nallapati et al., 2016; Durrett et al., 2016).

By emphasizing a query-based attention mechanism, Nema et al. (2017) implemented a diversity-driven model that reduces repetitive phrases in summaries. Abdullah and Chali (2020) proposed a query generation approach that considers both the input document and the target summary. Similarly, Xu and Lapata (2020) addressed the problem of query-cluster interaction and proposed a coarse-to-fine model for query-focused multi-document extractive summarization.

In this paper, we propose an evidence-based model that leverages a transfer learning approach to generate evidence-based queries. First, we train a model to generate evidence keywords from article, highlight pairs in the CNN/DailyMail dataset. We then use this evidence model to generate evidence-based queries for the Debatepedia and TD-QFS datasets.

3 Problem Definition

Given a query Q_i and a document D_i , we generate a query-relevant summary S_i in the QFS task. A query should focus on the parts of the document that are related to its keywords, while the summary should cover the corresponding query-relevant contexts within the document. We define the common context words present in both the document and the summary as evidence in this work. However, the challenge lies in generating such evidence using only the document. Therefore, we hypothesize that a transfer learning approach can help train an evidence-based model for the document-to-evidence-based query generation task.

4 Our Framework

In query-focused summarization, the summary should align with the query, meaning that query-related information must be present in the document. Therefore, the query should be supported by both the summary and the document, implying that evidence keywords must be reflected in the query. However, only a few QFS datasets provide query-document-summary triads, and these are often constructed based on simplifying assumptions. For example, in the Debatepedia dataset, questions from controversial debates are treated as queries, while topic titles are treated as summaries. In such cases, some titles (summaries) may not be fully relevant to the queries.

Motivated by this limitation, we investigate whether evidence keywords can perform better than

the original queries in QFS datasets. If our hypothesis holds, evidence-based queries could be applied to query-free resources for QFS tasks. Therefore, we propose an evidence-based model that generates evidence from documents to be used as queries. Our work¹ consists of two main steps. First, we fine-tune a pre-trained model on the CNN/DailyMail dataset for the document-to-query generation task. Then, using the evidence-based model, we generate evidence-based queries for the Debatepedia and TD-QFS datasets without accessing the summaries. This transfer learning approach helps us avoid target leakage while generating evidence-based queries.

We use the CNN/DailyMail dataset, which contains news article-highlight pairs, to generate evidence-based queries. Specifically, we extract the common words from news articles and their corresponding highlights and consider them as evidence using Equation 1:

$$E_i \leftarrow \{w_{ij}\} \text{ (if } w_{ij} = w_{ik}) \quad (1)$$

where E_i is the set of extracted evidence keywords from the news article (N_i) and the highlight (H_i) of the i th sample. Here, w_{ij} are the tokenized words from N_i , and w_{ik} are the tokenized words from H_i .

The T5 model has been successfully applied to various downstream tasks; therefore, we select it as our pre-trained model and fine-tune it for the evidence generation task. The news articles are fed into the encoder, and the extracted evidence is provided to the decoder for supervised fine-tuning. We use the cross-entropy loss function to update the model parameters during backpropagation. The overall architecture of our evidence-based model is illustrated in Figure 1.

5 Evaluation Detail

5.1 Intrinsic Evaluation

To determine the similarity between our evidence-based queries and the original queries, we performed an intrinsic evaluation using the open-source library spaCy².

¹Our code is available at <https://anonymous.4open.science/r/Our-Project>

²<https://spacy.io/>

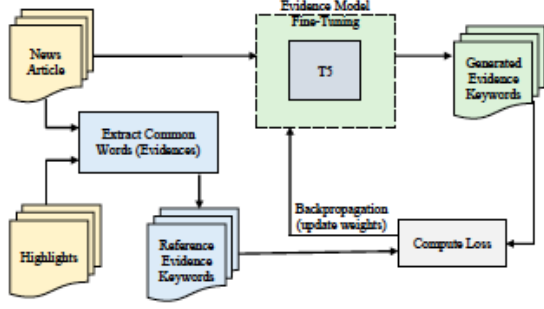


Figure 1: Evidence Model - Fine-tuning T5 in CNN/DM (News articles, Highlights)

5.2 Extrinsic Evaluation

5.2.1 Summarization using Pre-trained Models

After generating the evidence-based queries, we ranked the sentences in each document according to their relevance to the generated queries, thereby transforming the documents into query-relevant inputs. This ranking ensures that query-related sentences appear at the beginning of the document, reducing the risk that important information is truncated due to input size limitations. Finally, we fine-tuned several pre-trained summarization models (four models were used separately in our experiments) on the query-relevant documents from the Debatedpedia dataset to generate query-focused summaries.

Sentence Ranking

We used the Debatedpedia dataset for our QFS task, where sentence ranking helps prepare documents as query-relevant text inputs. First, we applied the evidence-based model to generate evidence-based queries for the documents in the Debatedpedia dataset. Then, for each sample, we split the document into a list of sentences and converted all texts—including the generated evidence-based query and the document sentences—into their corresponding vector representations using Equations 2, 3, and 4, respectively. Next, we computed the similarity between each sentence and the query using spaCy’s similarity metric, as shown in Equation 5. Finally, we sorted all sentences in descending order of their similarity scores to construct a query-relevant document, as described in Equation 6.

$$S_i = \text{sentenceTokenization}(D_i) \quad (2)$$

$$E_i^{vec} = \text{Doc2Vec}(E_i) \quad (3)$$

$$s_j^{vec} = \text{Doc2Vec}(s_j); [s_j \in S_i] \quad (4)$$

$$s_j^{sim} = \text{spaCy.similarity}(E_i^{vec}, s_j^{vec}) \quad (5)$$

$$D_i^E = \{s_1, s_2, \dots, s_p, s_q, \dots, s_{|D_i|}\} \quad (6)$$

where S_i is the list of sentences of the document D_i in the i^{th} sample. E_i^{vec} and s_j^{vec} are the vector representations of the evidence E_i and the sentence s_j , respectively. s_j^{sim} is the similarity score between evidence-based query and the j^{th} sentence of the document. Finally, D_i^E represents the query focused document of the i^{th} sample where $[\forall p, q \ s_p^{sim} \geq s_q^{sim}, p < q]$

Summarization Model

We used transformer-based pre-trained models for our summarization task: PEGASUS, BART, RoBERTa, and LED. Since these models can handle a limited number of input tokens (1024 tokens for PEGASUS, BART, and LED, and 514 tokens for RoBERTa), it is important to place the most query-relevant tokens at the beginning of the input sequence during training to generate query-focused summaries effectively. Our sentence-ranking approach ensures that the most query-relevant sentences appear at the beginning of the input.

Pre-trained models are typically trained on specific downstream tasks and can be fine-tuned for similar tasks on different datasets. We selected PEGASUS, BART, RoBERTa, and LED because they are pre-trained for summarization or sentence generation tasks. We then fine-tuned these models on query-focused documents from the Debatedpedia dataset for the QFS task to evaluate our hypotheses.

5.2.2 Summarization using a SOTA QFS Model

We conducted another experiment using QuerySum (Xu and Lapata, 2020), a recent state-of-the-art (SOTA) QFS model, to compare the results obtained using the original queries and the evidence-based queries on the TD-QFS dataset.

6 Experimental Setup

6.1 Datasets

We used the CNN/DailyMail dataset to train the evidence-based model, and the Debatedpedia and TD-QFS datasets to evaluate it using both the generated evidence-based queries and the original queries.

Dataset	Similarity Score
Debatepedia	0.69
TD-QFS	0.87

Table 2: Similarity score between original query and the evidence-based query

6.2 Implementation Details

We used 70K training and 1,337 validation samples from the CNN/DailyMail dataset to fine-tune the evidence-based model. For the summarization task, we used 12K training and 719 validation samples from the Debatepedia dataset, along with five pre-trained models: T5, PEGASUS, BART, RoBERTa, and LED.

We used a similar parameter configuration to fine-tune both the evidence generation and summarization models. We set the number of epochs to 3, weight decay to 0.01, and learning rate to $5e-05$. We used the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1e^{-08}$. The training batch size was set to 8, and the evaluation batch size to 32. During fine-tuning of the evidence-based model, we set warmup steps to 5,000 and evaluated the model every 500 steps. For the summarization models, we set warmup steps to 1,000 and evaluated the models every 250 steps.

To implement the QuerySum model, we followed the instructions provided by [Xu and Lapata \(2020\)](#) for both experimental settings, using original and evidence-based queries, respectively.

7 Results and Discussion

After performing the intrinsic evaluation, we computed the similarity scores between the original queries and the evidence-based queries for the Debatepedia and TD-QFS datasets, as shown in Table 2. The Debatepedia dataset yields a lower similarity score since its queries are formulated as questions. In contrast, both TD-QFS queries and our evidence-based queries are represented as sets of keywords.

We did not aim to achieve state-of-the-art results on the Debatepedia dataset for the QFS task; rather, our goal was to demonstrate that evidence-based queries can perform better than the original queries available in the dataset across four different pre-trained models. Our experimental results are shown in Table 3.

From Table 3, we observe that our evidence-based queries consistently outperform the original

MODEL	Query	R-1	R-2	R-L
Pegasus	Original	25.75	9.64	23.70
	Evidence	26.35	9.71	24.10
BART	Original	39.26	24.87	37.81
	Evidence	40.41	26.03	39.12
RoBERTa	Original	19.35	2.78	17.81
	Evidence	19.55	3.09	18.22
LED	Original	37.73	22.34	36.13
	Evidence	38.08	22.41	36.48

Table 3: Performance of Pegasus, BART, RoBERTa and LED models on Debatepedia dataset. R-1, R-2 and R-L stand for the F1 score of ROUGE 1, 2, and L, respectively

Query	R-1	R-2	R-SU4
Original	36.22	10.29	15.19
Evidence	35.95	9.97	15.94

Table 4: Performance using QuerySum ([Xu and Lapata, 2020](#)) model on TD-QFS dataset. R-1, R-2 and R-SU4 stand for the F1 score of ROUGE 1, 2, and SU4, respectively

dataset queries in generating summaries across all four pre-trained models. BART achieves the highest ROUGE-1, ROUGE-2, and ROUGE-L scores among the four models. LED shows the second-best performance, while RoBERTa obtains the lowest scores, except for the precision values of ROUGE-1 and ROUGE-L.

By implementing the same setup as the QuerySum model, we obtained ROUGE scores that differ from those reported in the original paper. Therefore, we report the results obtained from our own experimental environment. After replacing the original queries with evidence-based queries, we achieved a higher ROUGE-SU4 score, while ROUGE-1 and ROUGE-2 remained very close to the original scores. These results are presented in Table 4.

Based on the results in Tables 3 and 4, we conclude that our evidence-based model successfully replaces the original queries in the Debatepedia and TD-QFS datasets for the QFS task, achieving improved performance. Hence, this evidence-based model can support query-free summarization datasets by generating queries from their documents for the QFS task.

8 Conclusion

In this work, we present an evidence-based query generation model and provide comparative evidence that our approach successfully helps summarization models generate better summaries for query-focused datasets. In the future, we would like to extend our query generation approach to large-scale query-free datasets and further investigate how the generated queries support the QFS task.

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A Appendix

In Table 5, we are showing the generated queries and the original queries of the TD-QFS data set, where sample number 0 2 denotes the sample belongs to document number 2 of cluster number 0.

Sample Number	Original Query	Evidence-based Query
0 0	asthma causes	asthma chronic disease affects air
0 1	asthma treatment	asthma chronic lung disease inflam
0 2	exercise induced asthma	exercise induced asthma relatively common
0 3	atopic dermatitis	atopic dermatiti
0 4	atopic dermatitis	atopic dermatiti
0 5	asthma allergy	asthma chronic lung disease inflam
0 6	asthma medication	right medications asthma depend number things
0 7	exercise for asthmatic	exercise help prevent asthma attacks control
0 8	atopic dermatitis medications	atopic dermatiti
1 0	salt obesity	salt intake cause obesity people adults
1 1	obesity screening	body mass index measure used determine
1 2	childhood obesity	childhood obesity serious medical condition affect
1 3	causes of childhood obesity	children worldwide either overweight obese there
1 4	obesity and lifestyle change	obesity become increasing epidemic affects
1 5	obesity and diabetes	almost people type diabetes overweight number
1 6	retaining fluid water	retention also known medical term
1 7	body mass index	body mass index simple mathematical formula
1 8	childhood obesity low income	childhood obesity rates increased percent
1 9	obesity and nutrition	americans eating foods high calories
1 10	obesity metabolism	researchers mice born without section gene
1 11	emergence of type 2 diabetes	type diabetes caused combination factors including
2 0	Lung cancer	there basically types lung cancer there
2 1	Lung cancer diagnosis	lung cancer screening currently routine practice
2 2	Stage of lung cancer	stage refers extent lung cancer
2 3	Types of lung cancer	scle accounts
2 4	Lung Cancer in Women	smoking contributes lung cancer cigarettes
2 5	Lung Cancer chemotherapy	chemotherapy main treatment small cell lung
2 6	Lung cancer causes	lung cancer uncontrolled cell growth
2 7	Lung cancer treatment	lung cancer forms tissues usually cells
2 8	Lung cancer staging tests	staging helps doctor decide kind treatment
2 9	Non-small cell lung cancer	treatment non-small cell lung cancer
2 10	Risk factors for Lung Cancer in Women	women account lung cancer cases women
3 0	alzheimer memory	alzheimer common form
3 1	cognitive impairment	mild cognitive impairment mild cognitive impairment
3 2	alzheimer's symptoms	alzheimer symptoms include
3 3	semantic dementia	semantic dementia progressive neurodegenerative
3 4	helping retrieve memory alzheimer	experimental drug could help recover memory
3 5	Vascular Dementia	vascular dementia affects different
3 6	alzheimer diagnosis	alzheimer considered disease
3 7	first symptoms dementia	dementia specific disease describes wide range

Table 5: Original and the evidence-based queries from TD-QFS dataset, where sample number 0 2 indicates that the sample belongs to document number 2 in cluster number 0.